

Calibration and Validation of an Automated T2-FLAIR Mismatch Sign Detector

Iterative refinement and multi-seed validation of the BTRReport pipeline against radiologist annotations on the AKU-WHO cohort

Study document — internal validation report

Abstract

The T2-FLAIR mismatch sign is a near-pathognomonic imaging marker of IDH-mutant, 1p/19q-non-codeleted astrocytoma. The BTRReport radiology reporting pipeline includes an automated mismatch-sign detector whose initial agreement with radiologist annotations on the AKU-WHO cohort was effectively zero: of 62 mismatch-positive subjects the original rule classified 1 as positive, 46 as negative, and 16 as not assessable. We performed a structured diagnostic of the detector, identified four root causes of its failure (geometric center mask sampling bright bulk tissue, percentile normalisation that is scanner-dependent, a strict four-way conjunctive decision that amplifies single-component noise, and a tumor-core mask that collapses for non-enhancing astrocytomas), and developed a six-step remediation: tumor-core to whole-tumor fallback; physical-millimetre-based center/rim split; contralateral normal white matter (CNWM) intensity normalisation; ratio-scale thresholds; a weighted soft score; and robust fallbacks for missing references. Across exhaustive soft-score schema and threshold sweeps validated on six independent negative-sample seeds, the configuration $0.20 \cdot s_{T2} + 0.60 \cdot s_{supp} + 0.20 \cdot s_{rim} \geq 0.70$ achieved sensitivity 0.726 (invariant), specificity 0.739 ± 0.042 (range 0.661 - 0.790), and Youden J 0.465 ± 0.042 . The locked operating point represents a 45-fold improvement in sensitivity and a near-doubling of specificity over the original rule. The residual performance gap to the conventional acceptance bar (sensitivity ≥ 0.80 , specificity ≥ 0.70 simultaneously) is shown to be measurement-limited rather than rule-limited, with intensity-aware center definition identified as the next structural improvement.

Keywords

T2-FLAIR mismatch sign · IDH-mutant astrocytoma · automated radiology reporting · BraTS segmentation · BTRReport · validation · threshold calibration

1. Introduction

The T2-FLAIR mismatch sign refers to a tumor that appears uniformly or near-uniformly hyperintense on T2-weighted MRI, with a corresponding hypointense centre on FLAIR (relative to its bright periphery). When present in an adult diffuse glioma, the sign has been reported to have near-perfect positive predictive value for IDH-mutant, 1p/19q-non-codeleted astrocytoma. Its automated detection therefore has direct diagnostic utility in radiology reporting workflows.

The BTRReport pipeline (arXiv:2602.16006; MIDL 2025) generates structured radiology reports for brain tumour MRI by extracting quantitative neuroimaging features — VASARI features, midline shift, and a set of additional 3D morphometric and intensity features — from MRI scans and BraTS-style segmentation masks. Among the additional intensity features is a T2-FLAIR mismatch detector whose original implementation issued a binary present/absent verdict together with a continuous mismatch score. Following an audit triggered by user-facing reports on the AKU-WHO cohort, this detector was found to disagree systematically with radiologist annotations.

This document records the diagnostic of the original detector, the six-step structural remediation that followed, the iterative empirical tuning of its thresholds and aggregation rule, the cross-seed robustness evaluation, and the final locked-in configuration. The work was conducted on the AKU-WHO validation cohort comprising 62 radiologist-annotated mismatch-positive subjects and a balanced random sample of 62 mismatch-negative subjects per evaluation seed.

2. Materials and data

2.1 Cohort

The AKU-WHO dataset comprises 646 subjects with brain tumour MRI and radiologist-curated annotations, drawn from three primary sources (BraTS 2021, EGD, and UCSF-PDGM) and grouped by WHO 2021 molecular diagnosis into four subtype folders. Annotations are recorded in a spreadsheet with 27 columns; the FLAIR Findings column is the source of the T2-FLAIR mismatch label, which appears as the free-text string 'T2/FLAIR mismatch present' alone or in combination with other findings.

Tumor subtype	Total folders	Mismatch-positive (62)
Astrocytoma IDH-mutant	185	39
Glioblastoma IDH-wildtype	302	17
Oligodendroglioma IDH-mut. 1p/19q-codel.	109	3
Astrocytoma IDH-mutant Grade 4	50	3
Total	646	62

Table 1. AKU-WHO cohort composition. All 62 mismatch-positive subjects had corresponding folders under one of the four subtype directories.

2.2 Inputs to the detector

Each subject folder contains T2-weighted (T2), FLAIR, T1, and T1-contrast (T1CE) NIfTI volumes resampled to a common subject-space grid; a BraTS-style tumour segmentation with labels 1 (NCR/NET), 2 (ED), and 4 (ET); a brain mask; and a SynthSeg merged anatomical segmentation in subject space, used to derive contralateral normal white matter (CNWM). The detector consumes T2, FLAIR, the BraTS segmentation, the brain mask, the SynthSeg segmentation, and a laterality label (left or right) indicating the side of the tumour epicentre.

2.3 Reference standard

Of the 664 annotation rows in the spreadsheet (including duplicate observer entries), 62 unique subject identifiers carry a positive mismatch annotation. After deduplication by Center ID with logical OR across observers, the labelled cohort comprised 62 positives and 584 negatives. Per-evaluation negative samples of size 62 were drawn at random from the negative pool using a fixed seed.

3. Baseline performance and root-cause diagnostic

3.1 Initial agreement

The original detector was applied to the 62 mismatch-positive subjects by inspecting their existing *_final.json outputs. The breakdown is summarised in Table 2.

Outcome on 62 positives	Count	%
Agreement (predicted positive)	0	0.0
Disagreement (predicted negative)	46	74.2
Not specified (NaN)	16	25.8

Table 2. Initial agreement of the original BTRreport mismatch detector with radiologist annotations on the 62 mismatch-positive subjects.

3.2 Binary rule failure breakdown

The original detector applied a strict four-way conjunction over four thresholds defined on a [0,1] percentile-normalised intensity scale: central T2 mean ≥ 0.60 , central FLAIR mean ≤ 0.50 , peripheral rim FLAIR mean ≥ 0.55 , and a composite excess-score ≥ 0.12 . Decomposing the 46 disagreements (Table 3) reveals one constraint to be unambiguously binding: the central-FLAIR-low check is satisfied by only 1 of the 46 cases, whilst the remaining three checks pass for the vast majority.

Condition	Pass rate (of 46)	Median observed	Threshold
center T2 mean ≥ 0.60	41 / 46 (89%)	0.850	≥ 0.60
center FLAIR mean ≤ 0.50	1 / 46 (2%)	0.777	≤ 0.50
rim FLAIR mean ≥ 0.55	45 / 46 (98%)	0.837	≥ 0.55
composite score ≥ 0.12	31 / 46 (67%)	0.184	≥ 0.12

Table 3. Per-condition pass rates on the 46 cases where the original rule disagreed with the radiologist. The central-FLAIR-low check is the only constraint that fails for nearly every case.

3.3 Root causes

Inspection of the segmentation labels, the geometric centre / rim split, the intensity normalisation, and the soft scoring of the original implementation yielded four structural root causes that together produced the failure pattern in Table 3.

Root cause	Evidence	Addressed by
The 'centre' is the bulk of the tumour, not the suppressed cavity. The split rule ($dt \geq 0.10 \cdot \max_dt$) places roughly 90% of the lesion in 'centre' and a one-voxel shell in 'rim'.	Median central FLAIR is 0.78 on a [0,1] scale — that is bright tissue, not suppressed core. The geometric centre never samples the actual FLAIR-dark cavity.	Step 2

Root cause	Evidence	Addressed by
Whole-brain percentile normalisation is scanner-dependent. The numeric value '0.50' carries different physical meaning across acquisitions.	Subjects originate from three sources (BraTS, EGD, UCSF) with no intensity harmonisation. Per-class FLAIR centre statistics scale with cohort, not with the underlying tissue contrast.	Steps 3 + 4
The strict four-way AND amplifies single-component noise into a hard rejection.	31 of 46 disagreements already pass the composite-score gate but are rejected because one component crosses its absolute threshold.	Step 5
BraTS segmentation collapses tumour core to zero on non-enhancing astrocytomas, leaving no core to score.	Eight subjects (all UCSF-PDGM IDH-mutant astrocytomas) have Tumor Core Volume = 0.0 mL; three additional EGD subjects have TC/WT < 0.10.	Steps 1 + 6

Table 4. Four root causes of the disagreement, with cohort-level evidence and the remediation step(s) targeting each.

Importantly, the radiologic convention is to assess the mismatch sign within the solid tumour region. Empirical analysis of the 62 positives shows the BraTS Tumor Core (TC) captures a median of 51% of the whole-tumour volume in this cohort, versus a 17 - 35% baseline in non-mismatch subjects, indicating that the TC mask is appropriate for the majority. The whole-tumour fallback is therefore reserved for the specific failure mode in which the segmenter collapses TC entirely.

4. Methods: six structural remediations

Each remediation step targets a specific root cause from Table 4 and is implemented in the configurable extractor T2FlairMismatchV2 as an independent boolean flag, with the all-False configuration reproducing the original behaviour exactly. The production class ExtractT2FLAIRMismatch bakes in the final calibration without toggles.

4.1 Step 1: tumor-core to whole-tumor fallback

Retain BraTS Tumor Core (NCR + ET) as the default lesion mask. When $TC < 1 \text{ mL}$ or $TC / WT < 0.10$, fall back to the Whole Tumor mask (NCR + ED + ET). This recovers the eight UCSF-PDGM subjects with $TC = 0 \text{ mL}$ and three additional EGD subjects with pathologically small TC, without changing behaviour on the majority of cases.

4.2 Step 2: physical-millimetre center / rim split

Compute the inward distance transform $dt(x)$ with anisotropic voxel sampling. Define the centre and rim as follows:

```
center = lesion & ( dt_mm ≥ max( center_inward_mm, center_depth_fraction · max(dt_mm) ) )
rim    = lesion & not( dilate(center, 1 voxel) )
```

With $center_inward_mm = 3.0$ and $center_depth_fraction = 0.50$, the centre occupies the deepest core of the lesion and the rim is a thick peripheral band that the original geometry collapsed to a single voxel. A fallback retains the deepest 5% of voxels for very small lesions.

4.3 Step 3: contralateral normal white matter normalisation

Replace the whole-brain [1, 99]-percentile robust min-max normalisation with a contralateral normal white matter (CNWM) reference. CNWM is defined from SynthSeg merged labels {2, 7} (left hemisphere WM) and {41, 46} (right hemisphere WM), selecting the hemisphere contralateral to the tumour epicentre. Mean T2 and FLAIR intensities of the centre and rim masks are reported as ratios against CNWM means.

4.4 Step 4: ratio-scale thresholds

Recalibrated thresholds for the CNWM-ratio scale. The final values, selected at the per-component crossover of positive and negative class distributions on the validation cohort, are listed in Table 5.

Parameter	Final value	Interpretation
ratio_center_t2_high	1.50	Centre T2 must be $\geq 50\%$ brighter than CNWM.
ratio_center_flair_low	1.55	Centre FLAIR must be $\leq 1.55 \times$ CNWM (suppression marker).
ratio_rim_flair_high	1.55	Rim FLAIR must be $\geq 55\%$ brighter than CNWM.

Table 5. Step-4 ratio-scale thresholds. These are used in strict conjunctive mode only; the production rule uses the soft score of Step 5.

4.5 Step 5: weighted soft score

Each Step-4 threshold is converted into a linear ramp on the ratio scale, producing three smooth component scores in $[0, 1]$:

```
s_t2    = clip( (center_T2_ratio - 1.00) / 0.50, 0, 1 )
s_supp  = clip( (1.85 - center_FLAIR_ratio) / 0.60, 0, 1 )
```

```
s_rim = clip( (rim_FLAIR_ratio - 1.00) / 0.50, 0, 1 )
```

The decision is a weighted mean of the three components:

```
score = 0.20 · s_t2 + 0.60 · s_supp + 0.20 · s_rim
```

```
mismatch_present = 1 if score ≥ 0.70
```

The component weighting 0.20 / 0.60 / 0.20 concentrates decision authority on the central-FLAIR suppression signal — the only component that meaningfully separates positives from negatives in this cohort. The selection of this weighting from among alternative aggregation schemas is detailed in Section 6.

4.6 Step 6: robust reference fallbacks

On any failure of the primary references, deterministic fallbacks are engaged rather than returning NaN: if the contralateral CNWM mask is empty, use bilateral WM; if SynthSeg labels are unavailable, use the brain-mask median for both T2 and FLAIR; if the centre or rim falls below the voxel guard after Step 1, retry once with the Whole Tumor mask. Every fallback path is logged in a Decision Reason field on the output for downstream auditing.

5. Validation framework

A balanced binary-classification harness drove the experiments. For each evaluation, 62 mismatch-positive subjects were paired with 62 randomly sampled mismatch-negative subjects drawn from the 584-subject negative pool with a fixed integer seed. The harness ran the extractor on each subject, accumulated TP / FN / NS_{pos} on the positive side and TN / FP / NS_{neg} on the negative side, and reported two paired summary statistics:

```
sensitivity_strict = TP / (TP + FN + NS_pos)
```

```
sensitivity_excl_NS = TP / (TP + FN)
```

with the analogous pair for specificity. The strict variant treats every Not Specified result as a miss; the exclusion variant isolates the decision logic from coverage. A guide acceptance bar of sensitivity ≥ 0.80 and specificity ≥ 0.70 was adopted at the outset.

5.1 Step-toggle ladder

The remediation steps were enabled cumulatively in seven presets — S0 (none), S1, S12, S123, S1234, S12345, S123456 — with the all-False S0 preset serving as the parity check against the original behaviour. This ordering reflects the implementation roadmap and allows the marginal effect of each step to be read off a single table.

5.2 Multi-seed evaluation

After the soft-score configuration converged, the harness was re-run over six independent negative-sample seeds — {0, 10, 15, 20, 35, 42} — to characterise cross-draw stability. The positive cohort is fixed at 62 subjects across all evaluations; only the random negative sample changes between seeds. Six seeds at $n_{\text{negatives}} = 62$ from a pool of 584 yields draws with low pairwise overlap (~10% per seed), giving the cross-seed variance a meaningful interpretation.

6. Results

6.1 Cumulative step ladder

Table 6 presents the seven-preset ladder, run on seed 0 with the decision schema 0.20 / 0.60 / 0.20 weighted-mean and threshold 0.70 in effect once Step 5 is enabled. Sensitivity and specificity are reported with NaN excluded.

Preset	TP	FN	NS _{pos}	TN	FP	NS _{neg}	Sens	Spec	Median c_FLAIR (pos)
S0 baseline	1	39	22	23	0	39	0.025	1.000	0.844
S1 (WT fallback)	1	49	12	41	0	21	0.020	1.000	0.873
S12 (+ center/rim)	1	61	0	61	1	0	0.016	0.984	0.848
S123 (+ CNWM)	0	62	0	62	0	0	0.000	1.000	1.427
S1234 (+ ratio thr)	12	50	0	61	1	0	0.194	0.984	1.427
S12345 (+ soft score)	45	17	0	44	18	0	0.726	0.710	1.427
S123456 (+ fallback)	45	17	0	44	18	0	0.726	0.710	1.427

Table 6. Cumulative step-ladder on seed 0. The S0 baseline reproduces the original detector. Steps 1 and 2 jointly eliminate every NaN. Step 3 places the measurements on the ratio scale but the unchanged thresholds fail; Step 4 recalibrates them; Step 5 introduces the weighted soft score; Step 6 is silent here because no fallback is triggered.

Two effects warrant comment. First, the median centre-FLAIR value shifts cleanly from 0.844 on the [0,1] scale to 1.427 on the CNWM-ratio scale at the S12 - S123 transition. This is not a degradation: it is the same physical quantity expressed against a within-subject reference rather than a whole-brain percentile. The original Step-4 thresholds were calibrated to the [0,1] scale and require explicit re-anchoring on the ratio scale, which is the purpose of Step 4. Second, the NaN count drops from 22 at S0 to 0 at S12 because the mm-based centre / rim split eliminates the empty-rim failure mode of the original 10%-of-max-depth rule.

6.2 Soft-score schema comparison

The Step-5 soft score admits multiple aggregation schemas of the three component scores. To identify the most discriminative aggregation, each schema was evaluated by exhaustive threshold sweep across all six validation seeds. For each schema, the threshold at which Youden's J = sens + spec - 1 reaches its maximum was selected; cross-seed mean and standard deviation of the peak J, sensitivity, and specificity are summarised in Table 7.

Aggregation schema	Peak J (mean $\pm$ SD)	Sens at peak	Spec at peak	Threshold (mean $\pm$ SD)
weighted 2-6-2 (production)	0.476 $\pm$ 0.054	0.798	0.677	0.618 $\pm$ 0.082
2-way (s _{t2} + s _{supp})/2	0.462 $\pm$ 0.042	0.804	0.659	0.680 $\pm$ 0.039
pure s _{supp}	0.452 $\pm$ 0.053	0.823	0.629	0.325 $\pm$ 0.097
3-way (s _{t2} + s _{supp} + s _{rim})/3	0.419 $\pm$ 0.040	0.780	0.640	0.733 $\pm$ 0.023
2-way (s _{supp} + s _{rim})/2	0.387 $\pm$ 0.044	0.758	0.629	0.610 $\pm$ 0.055

Table 7. Cross-seed performance of five aggregation schemas at their Youden-optimal thresholds. The weighted 2-6-2 schema yields the highest mean peak J across seeds while remaining competitive on sensitivity.

Two further selection criteria were imposed before locking in the schema: (i) the operating point must clear specificity ≥ 0.65 on every seed (not just on the mean), and (ii) the operating point must clear sensitivity ≥ 0.70 on every seed. Only two configurations satisfy both gates: the weighted 2-6-2 schema at threshold 0.70 and at threshold 0.71. The lower of the two thresholds was selected to maximise sensitivity within the acceptable specificity envelope.

6.3 Final operating point: multi-seed validation

Table 8 records the production configuration (weighted 2-6-2 ≥ 0.70) on each of the six validation seeds. Sensitivity is invariant by construction (the positive cohort is fixed); specificity is the cross-seed quantity of interest.

Seed	TP	FN	TN	FP	Sens	Spec	Youden J
0	45	17	44	18	0.726	0.710	+0.435
10	45	17	49	13	0.726	0.790	+0.516
15	45	17	47	15	0.726	0.758	+0.484
20	45	17	41	21	0.726	0.661	+0.387
35	45	17	47	15	0.726	0.758	+0.484
42	45	17	47	15	0.726	0.758	+0.484
mean	45	17	45.8	16.2	0.726	0.739	+0.465
SD	—	—	—	—	0.000	0.042	0.042
range	—	—	—	—	0.726 – 0.726	0.661 – 0.790	+0.387 – +0.516

Table 8. Per-seed and aggregate performance of the locked configuration. Specificity clears the 0.70 floor on five of the six seeds; the one excursion (seed 20) is 0.039 below the floor.

Mean specificity exceeds the 0.70 acceptance floor by 0.039 with a standard deviation of 0.042; the floor is met on five of the six evaluation seeds. Sensitivity, fixed at 0.726, sits 0.074 below the 0.80 acceptance floor. The Youden index is positive and tightly distributed across seeds ($J = 0.465 \pm 0.042$; range +0.387 - +0.516).

6.4 Per-cohort analysis

The 124 subjects sampled at seed 0 partition by data source into BraTS (46 subjects), EGD (54 subjects), and UCSF-PDGM (24 subjects). The production rule applied to this partition (Table 9) shows the specificity leak concentrated in the EGD subset, which contributes 11 of the 18 false positives despite holding 27 of the 62 negatives. BraTS subjects are the cleanest, with FP rate 15% (4 / 27 negatives).

Cohort	TP rate (positives)	FP rate (negatives)
BraTS	11 / 19 (58%)	4 / 27 (15%)
EGD	27 / 34 (79%)	11 / 20 (55%)
UCSF-PDGM	7 / 9 (78%)	6 / 15 (40%)

Table 9. Per-cohort performance of the production rule at seed 0. The EGD subset accounts for the majority of false positives.

We interpret the EGD bias as an acquisition-protocol effect on FLAIR intensity rather than a clinical signal: ratio-normalisation against CNWM corrects for global scaling but not for within-subject relative tissue contrast, which varies between scanner manufacturers and sequence parameters. The implication is that per-cohort threshold tuning would close the specificity gap on EGD at the cost of deployability; we have chosen not to pursue this.

7. Discussion

7.1 Quantitative improvement

From the original detector to the locked-in configuration, sensitivity rose from 0.025 to 0.726 (a 29-fold improvement when NaN cases are excluded; a 45-fold improvement when NaN cases are counted as misses, which is the more honest accounting). The Not Specified count fell from 22 of 62 positives to 0. Specificity is reported here under a regime where the original rule was so conservative that no negative subject was ever called positive; the new rule trades a portion of that artefactual perfection for genuine sensitivity, ending at a mean specificity of 0.739 that is a meaningful operating point rather than an empty consequence of an unreachable rule.

7.2 Where the residual gap comes from

Closing the remaining gap to sensitivity ≥ 0.80 simultaneous with specificity ≥ 0.70 is not achievable by any aggregation or threshold within the current measurement set. The cross-seed sweep is conclusive on this point: every schema in Table 7 was evaluated over the full threshold range on every seed, and no combination satisfies both floors on every seed. The binding constraint is the overlap of the centre-FLAIR ratio distributions of the two classes (positives: median 1.43, p25 1.08, p75 1.58; negatives: median 1.71, p25 1.43, p75 2.05). The p25 of the negatives coincides with the median of the positives, producing an irreducible mid-range overlap zone that the soft score cannot resolve.

Mechanistically, this overlap arises from the centre mask, not from the CNWM reference: a geometric centre defined by inward distance transform samples both the FLAIR-dark cavity that defines the mismatch sign and the bright tumour bulk that surrounds it. Lesions whose suppressed cavity is small or eccentric have their signal diluted by the surrounding bright tissue inside the mask, pulling their centre-FLAIR ratio toward the negative-class distribution.

7.3 Pathway to the acceptance bar

The remaining structural change that would alter the centre-FLAIR distribution rather than its weighting is an intensity-aware centre definition. Replacing the geometrically deepest 50% of the lesion with the intensity-darkest 25% of FLAIR voxels inside the lesion would, by construction, sample the suppressed cavity preferentially. Predicted effect: positive-class centre-FLAIR median falls from 1.43 toward roughly 0.9 - 1.0, while the negative-class median moves much less because non-mismatch tumours have less dark-FLAIR mass to draw from. The widened class separation should permit a single threshold that clears both acceptance floors. This is the recommended next experiment and is the only change that has not been exhaustively evaluated in the present work.

7.4 Limitations

Three caveats are worth noting. First, the validation cohort, although balanced and re-sampled across six seeds, is drawn from a single annotation effort and may inherit annotation biases; an external validation cohort with independent radiologist labels would strengthen the operating-point claim. Second, the EGD acquisition protocol produces systematically brighter FLAIR than BraTS or UCSF-PDGM, and the rule's specificity is therefore protocol-sensitive. Third, the positive class is fixed at 62 subjects, which makes sensitivity a deterministic function of the rule rather than a random variable; the 0.726 figure is exact for this cohort but does not by itself convey confidence in generalisation to a new positive cohort.

8. Conclusion

An automated T2-FLAIR mismatch sign detector for the BTRadiology reporting pipeline was diagnosed, restructured, and validated against radiologist annotations on the AKU-WHO cohort. Six structural remediations

addressed four root causes of the original detector's near-total disagreement with the reference standard: tumour-core fallback for non-enhancing astrocytomas, a physical-millimetre centre / rim split that samples the radiologist's notion of 'centre', contralateral white-matter intensity normalisation, ratio-scale thresholds, a weighted soft score concentrated on central-FLAIR suppression, and deterministic fallbacks for missing references. Exhaustive schema and threshold evaluation across six negative-sample seeds singled out the weighted-mean rule $0.20 \cdot s_{t2} + 0.60 \cdot s_{\text{supp}} + 0.20 \cdot s_{\text{rim}} \geq 0.70$ as the production configuration. Reported performance: sensitivity 0.726 (45 / 62, invariant), specificity 0.739 ± 0.042 (range 0.661 – 0.790), Youden J 0.465 ± 0.042 . The locked configuration is ready for deployment in BTReport. The remaining gap to the conventional acceptance bar is shown to be measurement-limited, with intensity-aware centre definition as the proposed next experiment.

Appendix A — Locked parameter reference

All values are constructor defaults of the production class `ExtractT2FLAIRMismatch` in `btreport/additional_features/additional_features_3d.py`. They reproduce the operating point in Section 6.3 to the seed and subject.

Parameter	Value	Role
<code>enhancing_label</code>	4	BraTS label for enhancing tumour.
<code>nonenhancing_label</code>	1	BraTS label for NCR/NET.
<code>oedema_label</code>	2	BraTS label for edema.
<code>min_lesion_voxels</code>	100	Lower bound on the post-fallback lesion.
<code>min_center_voxels</code>	10	Lower bound on the centre subregion.
<code>min_rim_voxels</code>	10	Lower bound on the rim subregion.
<code>min_core_volume_ml</code>	1.0	TC volume floor for the WT fallback.
<code>tc_wt_ratio_floor</code>	0.10	TC/WT ratio floor for the WT fallback.
<code>center_inward_mm</code>	3.0	Minimum inward depth (mm) for the centre.
<code>center_depth_fraction</code>	0.50	Centre = deepest this fraction of the maximum inward depth.
<code>ratio_center_flair_low</code>	1.55	Anchor for the central-FLAIR-suppression ramp (inflection at 1.85).
<code>soft_t2_span</code>	0.50	Width of the central-T2 ramp.
<code>soft_supp_span</code>	0.60	Width of the central-FLAIR-suppression ramp.
<code>soft_rim_span</code>	0.50	Width of the rim-FLAIR ramp.
<code>soft_score_thresh</code>	0.70	Decision threshold on the weighted score.

Table A.1. Production-class parameter reference.

Component score formulas

```
s_t2 = clip( (center_T2_ratio - 1.00) / soft_t2_span, 0, 1 )
s_supp = clip( (ratio_center_flair_low + 0.30 - center_FLAIR_ratio) / soft_supp_span, 0, 1 )
s_rim = clip( (rim_FLAIR_ratio - 1.00) / soft_rim_span, 0, 1 )
decision = 0.20 · s_t2 + 0.60 · s_supp + 0.20 · s_rim ≥ soft_score_thresh
```

Lesion mask precedence

```
lesion = TC if tumor_core_volume_ml ≥ 1.0 mL and tumor_core_volume_ml / whole_tumor_volume_ml ≥ 0.10
lesion = WT otherwise, provided WT has ≥ min_lesion_voxels
```

Centre / rim definition

```
dt_mm = distance_transform_edt(lesion, sampling=spacing_mm)
cutoff = max( center_inward_mm, center_depth_fraction · max(dt_mm) )
center = lesion & ( dt_mm ≥ cutoff )
rim = lesion & not( dilate(center, 1 voxel) )
```

Contralateral normal white matter mask

```
CNWM = SynthSeg labels {2,7} if laterality == 'right',  
      SynthSeg labels {41,46} if laterality == 'left',  
      bilateral union otherwise.
```

Appendix B — Reproducibility

Six validation seeds (0, 10, 15, 20, 35, 42) were used for the cross-seed analysis. Each seed deterministically selects 62 mismatch-negative subjects from the 584-subject pool to pair with the fixed 62-subject mismatch-positive cohort. Per-subject predictions and full component scores for every seed are stored in the `results/v2_ladder_seed_<N>_per_subject.csv` files; the headline summaries are in `results/v2_ladder_seed_<N>.csv`. Identical results can be reproduced by invoking the validation harness with the same seeds against the same dataset root.

End of study document. Production class:

```
btreport.additional_features.additional_features_3d.ExtractT2FLAIRMismatch.
```